

UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLUMBIA

UNITED STATES OF AMERICA,	)	
	)	
Plaintiff,	)	
	)	
v.	)	
	)	Civil Action No. 25-cv-3943
	)	
APPROXIMATELY 13.98 MILLION USDT	)	
	)	
	)	
Defendant.	)	
_____	)	

VERIFIED COMPLAINT FOR FORFEITURE *IN REM*

Plaintiff, the United States of America, through the U.S. Attorney for the District of Columbia, brings this verified complaint for forfeiture in a civil action *in rem* against approximately 13.98 million USDT,¹ hereinafter the “Defendant Property,” and alleges as follows:

JURISDICTION AND VENUE

1. This Court has original jurisdiction of this civil action by virtue of 28 U.S.C. § 1345, because it has been commenced by the United States, and by virtue of 28 U.S.C. § 1355(a), because it is an action for the recovery and enforcement of a forfeiture under an Act of Congress.
2. Venue is proper here under 18 U.S.C. § 3238.

¹ As explained below, USDT is a type of virtual currency called a stablecoin. USDT is “stable” because its price is pegged to the price of the U.S. dollar, meaning one USDT is generally equivalent to \$1. USDT is issued by a company called Tether.

NATURE OF THE ACTION AND STATUTORY BASIS FOR FORFEITURE

3. The United States files this *in rem* forfeiture action to seek forfeiture of the Defendant Property involved in, and constituting the proceeds of, wire fraud, wire fraud conspiracy, money laundering, money laundering conspiracy, computer fraud and computer fraud conspiracy in violation of 18 U.S.C. §§ 1030, 1343, 1349, 1956, and 371, respectively.

4. Procedures for this action are mandated by Rule G of the supplemental Rules for Admiralty or Maritime Claims and Asset Forfeiture Actions and, to the extent applicable, 18 U.S.C. §§ 981 and 983 and the Federal Rules of Civil Procedure.

5. Title 18 U.S.C. § 981(a)(1)(A) mandates forfeiture of any property, real or personal, involved in a transaction or attempted transaction in violation of 18 U.S.C. §§ 1956, 1957, or 1960, or any property traceable to such property.

6. Title 18 U.S.C. § 981(a)(1)(C) mandates forfeiture of property constituting or derived from proceeds traceable to, *inter alia*, any offense constituting “specified unlawful activity” as defined by 18 U.S.C. § 1956(c)(7), or a conspiracy to commit such offense. A violation of 18 U.S.C. §§ 1030 or 1343, or conspiracy to commit those offenses, constitutes specified unlawful activity under 18 U.S.C. § 1956(c)(7)(A) as an offense listed in 18 U.S.C. § 1961(1)(B).

7. Title 18 U.S.C. § 1030(a)(2) makes it a crime, *inter alia*, to intentionally access a computer without authorization and thereby obtain information from any protected computer. 18 U.S.C. § 1030(a)(4) makes it a crime, *inter alia*, to knowingly and with intent to defraud, access a protected computer without authorization, and by means of such conduct further the intended fraud and obtain anything of value. The term “protected computer” is defined in 18 U.S.C. § 1030(e)(2) and includes, *inter alia*, a computer used in or affecting interstate or foreign commerce or communication, including a computer located outside the United States that is used in a manner that affects interstate or foreign commerce or communication of the United States. *See Van Buren v.*

United States, 141 S. Ct. 1648, 1652 (2021) (definition of protected computer under 18 U.S.C. § 1030(e)(2)(B) includes “at a minimum . . . all computers that connect to the Internet”).

8. Title 18 U.S.C. § 1343 provides that whoever, having devised or intending to devise any scheme or artifice to defraud, or for obtaining money or property by means of false or fraudulent pretenses, representations, or promises, transmits or causes to be transmitted by means of wire, radio, television communication in interstate or foreign commerce, any writings, signs, signals, pictures, or sounds for the purpose of executing such scheme or artifice, commits the violation of wire fraud.

9. Title 18 U.S.C. § 1349 provides that whoever attempts or conspires to commit a violation of 18 U.S.C. § 1343 shall be subject to the same penalties as those prescribed for the offense, the commission of which was the object of the attempt or conspiracy.

10. Title 18 U.S.C. § 1956(a)(1)(A)(i) provides in relevant part that “whoever, knowing that the property involved in a financial transaction represents the proceeds of some form of unlawful activity, conducts or attempts to conduct such a financial transaction which in fact involves the proceeds of specified unlawful activity—with the intent to promote the carrying on of specified unlawful activity” is guilty promotional money laundering.

11. Title 18 U.S.C. § 1956(a)(1)(B)(i) provides in relevant part that “whoever, knowing that the property involved in a financial transaction represents the proceeds of some form of unlawful activity, conducts or attempts to conduct such a financial transaction which in fact involves the proceeds of specified unlawful activity— . . . knowing that the transaction is designed in whole or in part . . . to conceal or disguise the nature, the location, the source, the ownership, or the control of the proceeds of specified unlawful activity” is guilty concealment money laundering.

12. Title 18 U.S.C. § 1956(h) provides that any person who conspires to commit any offense of 1956 is subject to the same penalties as those prescribed for the offense the commission of which was the object of the conspiracy.

PROPERTY INFORMATION

13. The Defendant Property is virtual currency that the Federal Bureau of Investigation (“FBI”) lawfully seized pursuant to case number 25-sz-33 and then transferred to the custody of the United States Marshals Service (“USMS”). As explained herein, the Defendant Property consists of USDT, a stablecoin issued by the company Tether. Tether controls the computer code that creates/mints USDT tokens. Because Tether controls that computer code, it can freeze USDT tokens associated with particular types of virtual currency addresses. In this case, Tether voluntarily responded to a request from U.S. law enforcement by freezing the USDT associated with the virtual currency addresses listed below (collectively, the “Target Addresses”).² After receiving seizure warrant 25-sz-33, authorized by the Honorable Federal Magistrate Judge Zia M. Faruqui in the District of Columbia, Tether burned the USDT associated with the Target Addresses and reissued an equivalent amount of the USDT to the U.S. government. That reissued USDT is the Defendant Property. The following virtual currency addresses are the Target Addresses:

- a. TLq5jP2J2ebYghCX1mXP9gcFRfjPbcrfZs,
- b. TRkJ39s58grMSMqFN8YgToMxcaj95ENFxf,
- c. TDqroZaFsjWXhefyMqXthS2rvxzmWnavNZ,
- d. TBo6abebYprVWWABcXbpbgtgGRtmeRSrdz,
- e. TU4oX4BcjgfjQUFuKAC27rKzYvqS5hNyKH,
- f. TDzh3ss71yCVCdKqcuQjvgaguVkJtmWSFh,

² The number and value of USDT tokens seized in each Target Address is described further below.

- g. TEcpUeeiVqxmUujg2s5SAAtFxcdN6P9bVB1,
- h. TSSs4zyjdQM6L7a6v1umWmng3zzwzquSq8,
- i. TEZuPBkVcNyci5WM3zrzoqX6ho4hMXgnJ6,
- j. TNBWvFHbBDjd2JzPqwSPywh8rv1C4LXjdM,
- k. TQPQT3im4NW6MGZSQwJBXA2S6Yj2saSBSy,
- l. THSBNYfGL9ja7b5xU98f8ofU2iy2gXPjSN,
- m. TTi46bHBvmkgc8NcHLfLB2587XF7GfZfQs,
- n. TK6vKMXpG6Fsrt462Nahgzg8RuzLfemnRX,
- o. TZ8Rni5ixMx6SKVFY4Qz7QdZYnbtRuGCVR,
- p. TXwggJm8GMcbew9J6A5myt8Gi5dxM9j8ia,
- q. TQ48GtajcAVJmohSBvJ4rvC9oVNZuWz57q,
- r. TC6WxrikcxBQVN2vpChGwbJP9QEAPp7CoR,
- s. TNSn4yJoR96ZruiEWPUFLxLa1k73bVpkEh,
- t. TByiFBGTx83ptNhLiFcQJQxtnG66hs4GY3,
- u. TKvUXiHow912c8HPCpwrrdoz4z47NWwL9D,
- v. TDmBm5zUSiupoV71MXUittZqReN3h85nJc,
- w. TN7iF7MgBSBNYFr38b3jYiCUEbREJAiWjz,
- x. TG9ZAev1Q4335uM6Gwm5df9eFUFz8e6Xgo,
- y. TMYtjT2aGUtY16WKF1bkh4ssjoKFKiznqw,
- z. TLv2pzHZaHyY8k5nWfCE6cGGkLwdQCi3Dg,
- aa. TLDwBfJBRptX5nFh4rCbvhkhT5cGWbHfKR,
- bb. TWYCvyCmhikaVpKx6Zf3nGWUjYbVhSGQK5,
- cc. TRxz3DY9ELAoFSZyf4CuaLyEHhGYmQ8b2w,

dd. TTPjk6tJjHPWNxLwEGE4ZW8ADqGsd95zvc,
ee. TA9bzquWzBRKgjTPJ5gbWMWv86rYqc2yZj,
ff. TDh6QVjBaYF5kdN2qrREKydArESsG12N9h,
gg. TMunHrvu7UCM8Ro6c968SPwRx9EdLDnHW5,
hh. TPjsHN4KRdaWYvUbVdMUuAeL6k7qLpTTxW,
ii. TAZ6yaCrxyVED6634GKXPntR7v6aBn7Rb,
jj. TLVSDypgdtRTd1yuGNquCLMoDhgkVrCgmo,
kk. THhic3agJ4544GuJUhuEj7bixriYfkjb8,
ll. TTbgnFEFEemiculeXN3bvUE8fKbdx3di8Yb,
mm. TLivG1x3R7Wgjha1w7Xgz2XWcWRV82yK8c,
nn. TW6swH66FMUMgjxHHSUUy1qDKrXZxprAKV,
oo. TXKPga6EH25mexfQ7BMFGbVfGwz1rvqXVK,
pp. TDskBcHHaFYSJz3QuKBSd5Q9ZxEc8dPzyH,
qq. TXJeZYLn2Vr7DUwvrqceo5VRQ8b9EucxLn,
rr. TWr1G47DtfvPZ1QAH3RgV9c6SkSSe8a9hS,
ss. TKdBozGxEFNZoAwX3rYB9gsZ373jQAYgkm,
tt. TCxCHCndDkz2xNAaYibienjLaS8eZASaXu, and
uu. TVFhKSygmJA1Qu7hrC4eavg8Esp9AoH2pu.

STATEMENT OF FACTS

I. Background Regarding the FBI's Investigation into North Korean Cyber Actors

14. The FBI is investigating numerous virtual currency heists perpetrated by members of North Korean military hacking groups, known within the cybersecurity community as both the

“Lazarus Group/APT 38” and APT38.³ Since at least late-2014, the APT38 subjects have targeted banks and virtual currency exchanges, including banks and virtual currency exchanges located in the United States, and have successfully initiated fraudulent transfers at such banks and exchanges. From at least 2017 through 2025, the North Korean APT38 subjects have continued this targeting, and successfully conducted multiple virtual currency heists, netting hundreds of millions of dollars of virtual currency. For example, in or around August 2021, APT38 cyber actors stole approximately \$90 million⁴ in virtual currency from Japan-based virtual currency exchange Liquid. Then, in or around March 2022, APT38 cyber actors stole approximately \$615 million in virtual currency from foreign-based Sky Mavis. In or around June 2022, APT38 cyber actors stole approximately \$105 million in virtual currency from U.S.-based Harmony. Then, in or around November 2023, the APT38 cyber actors stole approximately \$107 million from HTX, formerly known as “Huobi.” The APT38 cyber actors who stole virtual currency from Liquid, Sky Mavis, Harmony, HTX, and others are part of an ongoing conspiracy, and have regularly used U.S.-based computer infrastructure to facilitate intrusions that are part of these virtual currency heists. As explained further below, the APT38 cyber actors used the networks/platforms of U.S.-based companies to facilitate the unauthorized intrusion into HTX’s systems.

³ APT or “Advanced Persistent Threat” is a term used to define and identify groups of organized, highly skilled, and well-resourced cyber actors who maintain focused efforts on specific tasks such as intelligence gathering against specific business sectors or governments. APTs are known to gain access to computer networks while remaining undetected for extended periods. APTs are often nation-state or state-sponsored groups. Upon identification, the group is assigned a unique number as an identifier by the community: in this case, the cybersecurity community has dubbed this group of North Korean cyber actors as “APT38.” In April 2022, the FBI, the Treasury Department, and DHS Cybersecurity and Infrastructure Security Agency issued a public Cybersecurity Advisory numbered AA22-108A which described specific APT38 tactics, techniques, and procedures targeting blockchain companies, naming this malicious North Korean cyber activity “TraderTraitor.”

⁴ The U.S. dollar conversions listed in this complaint are based on the rate at or around the time of the heist and/or at or around the time of the transaction.

15. The FBI assesses that North Korean “Lazarus” cyber actors more broadly have engaged in cyber-attacks, intrusions, and attempted intrusions into computers and networks of, among others, U.S. and foreign entertainment companies, U.S. and foreign banks, U.S. cleared defense contractors and energy companies, virtual currency exchanges, information security researchers, and pharmaceutical companies. The FBI also assesses that North Korean APT38 actors are responsible for certain hacks based on, among other things, distinctive tactics, techniques, and procedures observed in other virtual currency heists linked to North Korea. The North Korean APT38 actors have exhibited a particular, financially motivated focus on leveraging their malicious cyber activity to steal money and virtual currency from their victims. Private sector cybersecurity research companies, including, but not limited to, Google Cloud’s Mandiant in October 2023 and Wiz in July 2025,⁵ have corroborated the FBI’s assessment that APT38 is responsible for these heists.

16. As explained below, the Defendant Property is traceable to the HTX heist. The United States of America seeks to lawfully forfeit the Defendant Property to punish and deter criminal activity by depriving criminals of property used in or acquired through illegal activities, to promote and enhance cooperation among law enforcement agencies, and, most importantly, to recover assets that may be used to compensate victims through the remission process.⁶

⁵ “Assessed Cyber Structure and Alignments of North Korea in 2023,” <https://cloud.google.com/blog/topics/threat-intelligence/north-korea-cyber-structure-alignment-2023>, and “TraderTraitor: Deep Dive, Inside the Lazarus subgroup that’s hijacking cloud platforms, poisoning supply chains, and stealing billions in digital assets.” <https://www.wiz.io/blog/north-korean-tradertraitor-crypto-heist>

⁶ See United States Asset Forfeiture Program, *Our Mission*, <https://www.justice.gov/afp>.

II. Relevant Background Related to Virtual Currency

17. **Virtual Currency**: Virtual currencies are digital tokens of value circulated over the Internet. Virtual currencies are typically not issued by any government or bank like traditional fiat currencies, such as the U.S. dollar, but rather are generated and controlled through computer software. Different virtual currencies operate on different blockchains, and there are many different, widely used virtual currencies currently in circulation. Bitcoin (or BTC) and ether (ETH) are currently the most well-known virtual currencies in use. BTC exists on the Bitcoin blockchain, and ETH exists on the Ethereum network. Typically, a virtual currency that is “native” to a particular blockchain cannot be used on a different blockchain. Thus, absent technological solutions, those native assets are siloed within a specific blockchain. For instance, ETH (the native token on the Ethereum network) cannot be used on other networks unless it is “wrapped” by smart contract code. This wrapping process results in what is called Wrapped ETH or WETH.

18. **Stablecoins**: As mentioned above, stablecoins are a type of virtual currency whose value is pegged to a commodity’s price, such as gold, or to a fiat currency, such as the U.S. dollar, or to a different virtual currency. Stablecoins achieve their price stability via collateralizations (backing) or through algorithmic mechanisms of buying and selling the reference asset or its derivatives.

19. **Tether (USDT)**: Tether is a company that manages the smart contracts (which are defined below) and the treasury (*i.e.*, the funds held in reserve) for USDT, a stablecoin pegged to the U.S. dollar. As previously stated, Tether can update its smart contracts to, among other things, freeze, burn, and reissue USDT associated with particular virtual currency addresses.

20. **Virtual Currency Address**: Virtual currency addresses are the particular virtual locations to which such currencies are sent and received. A virtual currency address is analogous to a bank account number and is represented as a string of letters and numbers.

21. **Private Key**: Each virtual currency address is controlled through the use of a unique corresponding private key, a cryptographic equivalent of a password, which is needed to access the address. Only the holder of an address's private key can authorize a transfer of virtual currency from that address to another address.

22. **Virtual Currency Wallet**: There are various types of virtual currency wallets, including software wallets, hardware wallets, and paper wallets. The virtual currency wallets at issue for the purposes of the instant complaint are software wallets (*i.e.*, a software application that interfaces with the virtual currency's specific blockchain and generates and stores a user's addresses and private keys). A virtual currency wallet allows users to store, send, and receive virtual currencies. A virtual currency wallet can hold many virtual currency addresses at the same time. Wallets that are hosted by third parties are referred to as "hosted wallets" because the third party retains a customer's funds until the customer is ready to transact with those funds. Conversely, wallets that allow users to exercise total, independent control over their funds are often called "unhosted" wallets.

23. **Blockchain**: Many virtual currencies publicly record all of their transactions on what is known as a blockchain. The blockchain is essentially a distributed public ledger, run by the decentralized network of computers, containing an immutable and historical record of every transaction utilizing that blockchain's technology. The blockchain can be updated multiple times per hour and records every virtual currency address that has ever received that virtual currency and maintains records of every transaction and all the known balances for each virtual currency address. There are different blockchains for different types of virtual currencies.

24. **Blockchain Explorer**: These explorers are online tools that operate as a blockchain search engine allowing users the ability to search for and review transactional data for any addresses

on a particular blockchain. A blockchain explorer is software that uses API⁷ and blockchain nodes to draw data from a blockchain and uses a database to arrange and present the data to a user in a searchable format.

25. **Smart Contracts**: Smart contracts are computer programs stored on a blockchain that run when predetermined conditions are met. Typically, they are used to automate the execution of an agreement so that all participants can be immediately certain of the outcome, without any intermediary's involvement. The Ethereum network is designed and functions based on smart contracts.

26. **Virtual Currency Bridge**: A blockchain bridge, otherwise known as a cross-chain bridge, connects two blockchains and allows users to send virtual currency from one chain to the other—for example, from the Bitcoin blockchain to the Ethereum network.

27. **Virtual Currency Exchanges (VCEs)**: VCEs are trading and/or storage platforms for virtual currencies, such as BTC and ETH. There are generally two types of VCEs: centralized exchanges and decentralized exchanges, which are also known as “DEXs.” Many VCEs also store their customers' virtual currency in virtual currency wallets. As previously stated, these wallets can hold multiple virtual currency addresses associated with a user on a VCE's network. Because VCEs act as money services businesses, they are legally required to conduct due diligence on their customers (*i.e.*, know-your-customer checks) and to have anti-money laundering programs in place (to the extent they operate and service customers in the United States).

28. **Virtual Currency Mixers**: Virtual currency mixers (also known as tumblers or mixing services) are software services that allow users, for a fee, to send virtual currency to

⁷ API stands for “application programming interface,” which is a set of definitions and protocols for building and integrating application software.

designated recipients in a manner designed to conceal and obfuscate the source of the virtual currency. Based on my training and experience, I know that virtual currency mixers are a common laundering tool used by North Korean cyber actors and their money laundering co-conspirators. As described below, Tornado Cash and CryptoMixer are virtual currency mixers that the actors used to launder the proceeds of their heists.

29. **Blockchain Analysis:** As previously stated, while the identity of a virtual currency address owner is generally anonymous, law enforcement can often identify the owner of a particular virtual currency address by analyzing the blockchain (*e.g.*, the Bitcoin blockchain). The analysis can also reveal additional addresses controlled by the same individual or entity. “For example, when an organization creates multiple [BTC] addresses, it will often combine its [BTC] addresses into a separate, central [BTC] address (*i.e.*, a ‘cluster’). It is possible to identify a ‘cluster’ of [BTC] addresses held by one organization by analyzing the [BTC] blockchain’s transaction history. Open-source tools and private software products can be used to analyze a transaction.” *United States v. Gratkowski*, 964 F.3d 307, 309 (5th Cir. 2020).

30. In addition to using publicly available blockchain explorers, law enforcement uses commercial services offered by several different blockchain-analysis companies to investigate virtual currency transactions. These companies analyze virtual currency blockchains and attempt to identify the individuals or groups involved in transactions. Through numerous unrelated investigations, law enforcement has found the information provided by these tools to be reliable.

III. Background Regarding the HTX Heist and Law Enforcement’s Attribution of the Heist to North Korea

Overview of the HTX Heist

31. HTX is a global virtual currency exchange. As part of its services, HTX developed its own blockchain called the “Huobi ECO Chain,” or “HECO.” The HECO Bridge connects the

HECO blockchain with the Ethereum blockchain. On or about November 22, 2023, North Korean cyber actors stole approximately \$107 million in virtual currency from HTX and the HTX HECO Bridge. As described further below, the North Korean cyber actors and/or their money laundering co-conspirators then laundered the stolen funds through virtual currency bridges, several BTC and ETH addresses, and virtual currency mixers before consolidating and depositing the stolen funds into, among other places, the Target Addresses.

Evidence Indicating that North Korean Cyber Actors Conducted the HTX Heist

32. Shortly after the heist, attorneys for HTX provided the FBI with indicators of compromise (“IOCs”) from the intrusion. These IOCs included domain names, IP addresses, malware sample filenames, spear-phishing accounts, and other attacker techniques uncovered during their investigation of the theft. The IOCs are consistent with APT38 malicious cyber activity. Specifically, the IOCs involved the following:

- a. On or about November 10, 2023, an APT38 cyber actor socially engineered⁸ an employee at HTX and convinced that person to download and run a malicious application from a GitHub repository on the employee’s work laptop. (GitHub is a U.S.-based company.) The GitHub repository was located at “Github.com/MillerSpace.” The project containing the malicious application was named “ChallengeCoin.”
- b. On or about November 13, 2023, the HTX employee’s compromised computer downloaded and launched an application named “CryptoOracle” from the same “Github.com/MillerSpace” repository. Between on or about November 11 and 22,

⁸ Social engineering, in this context, means convincing a victim through deceit to take an action—such as running an application—which may compromise the victim’s security or expose sensitive information.

2023, the HTX employee's compromised computer connected to IP address 72.11.157.199 multiple times. IP address 72.11.157.199 corresponded to the domain deckfinisher.com, which was previously flagged by a U.S.-based cybersecurity company, as a domain associated with the malware family "TwoPence/Electric." This malware family belongs to, and is used by, the APT 38 malicious cyber actors.

c. On or about November 17, 2023, the APT38 malicious cyber actors directly connected, without authorization, to one of HTX's servers from the employee's compromised computer. The APT38 malicious cyber actors gathered system information on the server and uploaded a backdoor⁹ to the server. The APT38 malicious cyber actors renamed the uploaded backdoor to "kwork." Next, the server connected to IP address 185.117.75.183, which corresponded to the domain chapterstray.com, and executed the malicious process "xenwatch" in order to obtain persistent control of the server.¹⁰ Between on or about November 17, 2023, and on or about November 22, 2023, the compromised server repeatedly connected to the following IP addresses and domains:

- IP address 51.254.53.24 and domain hyphenstylus.com;
- IP address 95.179.140.133 and domain capeshowman.com;
- IP address 38.180.24.126 and domain jazzlabrador.com; and
- IP address 185.213.23.137 and domain gamingbreath.com.

⁹ A backdoor in computing terms is a malicious software program or tool which allows remote access into a computer system by an attacker.

¹⁰ "Persistence" in computer security terminology refers to techniques and tools used by cyber actors to maintain continuous access to a compromised system or network.

Previously, the FBI had identified the domains chapterstray.com, hyphenstylus.com, jazzlabrador.com, and gamingbreath.com as domains utilized by APT38 malicious cyber actors, among several other domains.

- d. On or about November 22, 2023, the APT38 malicious cyber actors uploaded a backdoor to other parts of HTX's online hosting infrastructure. Shortly thereafter, the APT38 malicious cyber actors stole HTX's assets (worth approximately \$107 million in virtual currency at the time of the hack) via their access to HTX's internal network. After the theft, the APT38 malicious cyber actors deleted the backdoors from HTX's servers and deleted the resolutions of the domains to the IP addresses.

IV. Overview Regarding the FBI's Efforts to Trace Funds from the HTX Heist to the Target Addresses

33. The following overview section describes—at a high level—the money laundering that followed the heist. The laundering transactions are described in further detail in section V. The details associated with each transfer are not listed in this complaint as to include those lengthy details would make this complaint unwieldy and the information provided sufficiently establishes the Defendant Property's connection to a crime. However, detailed tracing graphs can be made available for the court's review should the court wish to see them.

34. As previously stated, on or about November 22, 2023, APT38 cyber actors stole virtual currency from HTX and the HECO Bridge. During the heist, the APT38 cyber actors were able to steal virtual currency from the Ethereum, Bitcoin, and TRON blockchains. This complaint focuses on the virtual currency that the APT38 cyber actors stole from the Ethereum blockchain. After the APT38 cyber actors stole HTX's virtual currency, the APT38 cyber actors (and/or their money laundering co-conspirators) converted the virtual currency stolen from the Ethereum

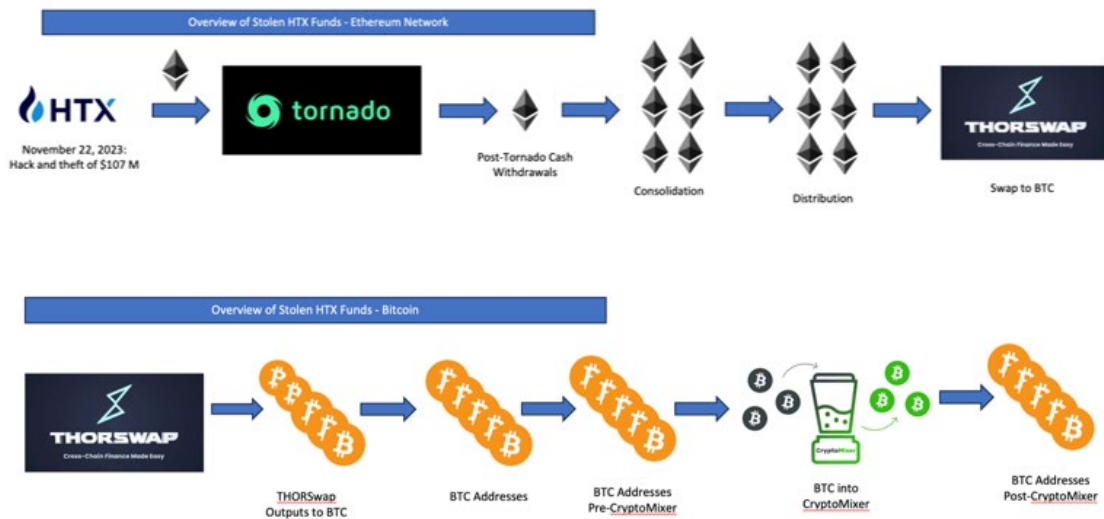
blockchain into ETH.¹¹ After ensuring all of the stolen virtual currency from the Ethereum blockchain was then held in ETH, the APT38 cyber actors (and/or their money laundering co-conspirators) transferred that ETH into Tornado Cash, a virtual currency mixing service.

35. After mixing the funds via Tornado Cash's platform, the APT38 cyber actors (and/or their money laundering co-conspirators) withdrew the funds from Tornado Cash. Those funds were still held in ETH but were now associated with different virtual currency addresses. Some of the ETH from these withdrawals went into THORSwap,¹² through which the APT38 cyber actors (and/or their money laundering co-conspirators) converted them into BTC. The BTC was then consolidated into batches of BTC and sent into CryptoMixer, another virtual currency mixing service. The use of mixers like Tornado Cash and CryptoMixer (in this fashion) is a money laundering technique used to obfuscate the true source of the transactions. As described below, the FBI and private sector partners have developed means of analysis to partially defeat these obfuscation techniques.

36. After the APT38 cyber actors (and/or their money laundering co-conspirators) sent the BTC through CryptoMixer, the FBI identified BTC consolidation points (i.e., virtual currency addresses that held chunks of the stolen funds). The flow of funds is summarized in the chart below. The black diamond logo is the symbol for ETH tokens. The logo with a "B" is a representation for BTC.

¹¹ For reference, one ETH traded for approximately \$2,063 on or about November 22, 2023.

¹² THORSwap is a decentralized virtual currency exchange that allows users to swap assets across various blockchains.



37. The BTC consolidations were then laundered through two pathways, as follows:
- The first pathway involved BTC sent to a specific consolidation BTC address, namely bc1qww0gm8c0d04snnfhnplkg9vla5uw7rnmwz5l0j (the “510j BTC Address”). From the 510j BTC Address, the first pathway included a portion of the stolen funds going through two virtual currency swapping services (i.e., a virtual currency service that allows a customer to exchange one type of virtual currency for another) and into some of the Target Addresses.

- b. The second pathway involved some of the BTC going into a peel chain¹³ and then into a cluster of addresses associated with BTC address bc1qasn7jp00fw85046yefwrypvdauag83ln59v764 (“v764 BTC Address”), which will be discussed further below. At this point in the laundering chain, the APT38 cyber actors (and/or their money laundering co-conspirators) utilized Wasabi Wallet’s CoinJoin¹⁴ mixing feature to mix the BTC before sending it through (1) other virtual currency swapping services (obfuscation that, as described below, the FBI and private sector partners are also able to partially defeat), (2) a consolidation address on the BNB Smart Chain (BSC)¹⁵, (3) another virtual currency swapping service, and (4) finally, into the remainder of the Target Addresses.

38. For ease of explanation, the theft and laundering activities outlined above can be broken down into three stages:

- a. **Stage 1** of the laundering involved the initial theft from HTX, the laundering of virtual assets on the Ethereum blockchain, and the conversion of the stolen ETH to BTC.
- b. **Stage 2** of the laundering involved moving the BTC through CryptoMixer and into multiple new BTC addresses referred to below as “post-CryptoMixer BTC.”

¹³ A peel chain occurs when a large amount of virtual currency sitting at one address is sent through a series of transactions in which a slightly smaller amount of virtual currency is transferred to a new address each time. In each transaction, some quantity of virtual currency “peel off” the chain to another address (frequently, to be deposited into a VCE), and the remaining balance is transferred to the next address in the chain.

¹⁴ Wasabi Wallet is an open-source Bitcoin wallet which utilizes the CoinJoin mixing feature which mixes user inputs and produces outputs in the same, repeated values to hide the user inputs.

¹⁵ The application filed on June 4, 2025, under 25-sz-33 in the US District Court for the District of Columbia inaccurately referred to this consolidation address as an Ethereum address, however this consolidation address was an address on the BNB Smart Chain, and was accurately described as such in the description portion of that application.

- c. **Stage 3** of the laundering involved the movement of post-CryptoMixer BTC into Wasabi Wallet or into the virtual currency exchange Bitget, then through swapping services, and finally converting stolen funds into USDT on the TRON blockchain (i.e., into the Target Addresses).

V. **Additional Details Regarding the FBI's Blockchain Tracing from HTX's Wallets to the Target Addresses**

Stage 1 - ETH Laundering and Conversion to BTC

39. As previously stated, on or about November 22, 2023, APT38 cyber actors hacked into HTX's networks and stole virtual currency from HTX that was valued at approximately \$107 million at the time of the heist. When the APT38 actors stole the funds, they were able to take virtual currency from the Ethereum, Bitcoin, and TRON blockchains. This complaint focuses on the stolen tokens that resided on the Ethereum blockchain, as it is a portion of those funds which were deposited into the Target Addresses. The APT38 actors (and/or their money laundering co-conspirators) converted those Ethereum-based tokens into ETH, for a total of approximately 48,194 ETH. Following this conversion, the stolen funds in ETH were transferred to newly created addresses, where the stolen funds (i.e., the 48,194 ETH) remained until on or about March 13, 2024.

40. On or about March 13, 2024, the APT38 actors started sending the stolen ETH into the Tornado Cash mixing service. Between on or about March 13, 2024, and on or about March 24, 2024, the actors made approximately 508 deposits of ETH to Tornado Cash. The majority of these transfers utilized the Tornado Cash 100 ETH contract,¹⁶ with the approximately 48,194 stolen ETH being transferred into Tornado Cash.¹⁷

41. After the APT38 actors (and/or their money laundering co-conspirators) funneled the stolen funds into Tornado Cash, they withdrew the funds out of Tornado Cash almost immediately (between on or about March 13, 2024, and on or about March 24, 2024), transferring the funds to 142 new ETH addresses. The 142 post-Tornado Cash ETH addresses held a total of approximately 47,981 ETH, nearly the same amount of ETH that had gone into Tornado Cash. The difference between the 48,194 and 47,981 ETH is likely due to fees collected for using Tornado Cash.

¹⁶ Because of the way Tornado Cash's smart contracts were written, all virtual currency (i.e., all ETH) sent to Tornado Cash needs to be sent in predetermined ETH amounts of 100, 10, 1, or .1. In this case, the actors sent ETH through Tornado Cash in increments of 100 (i.e., they used the Tornado Cash 100 ETH contract).

¹⁷ For reference, the price of one ETH varied from approximately \$4,000 to \$3,400 between on or about March 13, 2024, through on or about March 24, 2024.

42. Although mixing services like Tornado Cash are designed to obfuscate the trail of stolen funds, it is possible to use various methods to “demix” the mixing transactions. As previously stated, the APT38 actors use the 100 ETH Tornado Cash smart contract. Additionally, the APT38 actors (and/or their money laundering co-conspirators) commingled some of the ETH back together after they withdrew it from Tornado Cash. Because of that, agents could watch funds coming out of Tornado Cash to assess which funds were commingled with other APT38-derived funds. This allowed the FBI to determine that those outputs were a portion of the stolen funds. In other words, when those funds were withdrawn from Tornado Cash and then commingled with other funds known to have been stolen by the APT38 actors, it became clear that the funds were likely still in the hands of the actors and/or their money laundering co-conspirators.

43. During the course of this investigation, FBI personnel used commercial blockchain analysis tools to assist in identifying the 142 “post-Tornado Cash” ETH addresses. These tools assist in analysis of the amounts, withdrawal timing, structuring, and post-withdrawal consolidation of the stolen ETH. As previously stated, the 142 post-Tornado Cash ETH addresses that were identified held a total of approximately 47,981 ETH, nearly the same amount of ETH that had gone into Tornado Cash.¹⁸

¹⁸ To utilize Tornado Cash, a user will pay a “relayer fee,” which would account for the difference in the amount of ETH withdrawn from the amount of ETH deposited.

44. After being sent through Tornado Cash, between on or about March 20, 2024, and on or about March 27, 2024, the APT38 actors (and/or their money laundering co-conspirators) transferred 41,209 ETH (out of the 47,981 ETH referenced above) via 126 transfers to approximately 13 consolidation ETH addresses. From these 13 consolidation ETH addresses, the stolen funds (i.e., the approximately 41,209 ETH) were broken up further and distributed into 137¹⁹ ETH addresses which then sent funds into THORSwap.²⁰

Stage 2 – Actors Convert ETH to BTC and Launder via CryptoMixer

45. Next, as reflected in the publicly available transaction explorer referenced below, the actors used the THORSwap Bridge to convert the approximately 41,209 ETH into BTC. The resultant BTC consisted of approximately 124 BTC addresses that received approximately 2,401 BTC from THORSwap via approximately 745 deposits between on or about March 20, 2024, through on or about April 1, 2024.

46. Then, between on or about March 20, 2024, through on or about April 1, 2024, these 124 BTC addresses made approximately 718 transfers to 716 different BTC addresses, totaling approximately 2,374 BTC, all of which is traceable to the HTX heist.

¹⁹ In or around August 2024, the FBI applied for a separate seizure warrant tied to this HTX heist. That seizure warrant was directed at a different provider for different funds tied to the HTX heist. That prior seizure warrant affidavit was filed in the Central District of California under case number 24-mj-04835, as the FBI was initially investigating that heist with the U.S. Attorney's Office for the Central District of California. In that separate filing, FBI identified 136 ETH addresses sending funds into THORSwap. Additional tracing review after that warrant was filed led the FBI to identify one additional ETH address involved in this phase of the laundering. For that reason, the instant complaint references 137 addresses and not 136.

²⁰ As noted above, THORSwap is a decentralized virtual currency exchange which allows users to swap assets across various blockchains. THORSwap Bridge transactions, such as these ETH to BTC transactions, can be viewed on the Internet via the publicly available transaction explorer at <https://thorchain.net/>.

47. Next, between on or about March 20, 2024, through on or about April 3, 2024, these 716 BTC addresses made 733 additional transfers to approximately 539 different BTC addresses, totaling approximately 2,356 BTC. These 539 BTC addresses holding approximately 2,356 BTC comprised the BTC that was segregated, that is, separated out and staged in addresses waiting to be deposited) into CryptoMixer. This segregated/separated BTC is referred to herein as the “Pre-CryptoMixer BTC.” The APT38 actors commonly use several mixing services designed to obfuscate the source of the stolen funds, including CryptoMixer and Tornado Cash.

48. Between on or about March 22, 2024, through on or about April 3, 2024, the APT38 actors (and/or their money laundering co-conspirators) sent 2,125 out of the 2,356 in Pre-CryptoMixer BTC via 227 transfers into CryptoMixer. Law enforcement identified approximately 1,688 BTC addresses and 2,000 transactions (1,126 withdrawals and 874²¹ deposits) involved in the laundering process of the approximately 2,125 BTC within CryptoMixer.

²¹ As previously stated, in or around August 2024, the FBI applied for a separate seizure warrant tied to this HTX heist. That seizure warrant was directed at a different provider for different funds tied to the HTX heist. That prior seizure warrant affidavit was filed in the Central District of California under case number 24-mj-04835, as FBI was initially investigating that heist with the U.S. Attorney’s Office for the Central District of California. In that separate filing, FBI inadvertently identified “824” deposits instead of “874.” 874 is the correct number, which is why FBI updated that in this complaint.

49. As the APT38 actors (and/or their money laundering co-conspirators) withdrew funds from CryptoMixer, each transaction was reviewed and flagged by FBI as a potential candidate for laundered stolen funds, or potential “post-CryptoMixer BTC.” As part of their laundering process, the APT38 actors generally deposit and withdraw from mixing services very quickly and typically within less than one or two hours. The FBI assessed that the flagged funds were connected to the HTX heist based on, among other things, the fact that the post-CryptoMixer BTC was withdrawn from CryptoMixer to intermediary addresses (i.e., unhosted wallet addresses that appear to only be used for obfuscating the source and destination of transfers) with the same transaction types and wallet details. By noting these characteristics and following these post-mixing consolidation points, the FBI was able to trace the flow of the stolen funds through CryptoMixer to the third stage of the APT38 actors’ laundering process.

Stage 3 – Post-mix Consolidation and Transfer to the Target Addresses

50. Between on or about March 22, 2024, and on or about March 28, 2024, some of the post-CryptoMixer BTC, that is, BTC traced as stolen funds from the HTX hack, went into approximately 268 BTC addresses via 307 withdrawals.²² Approximately 1,093 BTC (all of which is connected to the HTX heist) was transferred to the 268 BTC addresses from CryptoMixer. Notably, between on or about March 22, 2024, and on or about March 25, 2024, these 268 BTC addresses also received approximately 52.5 BTC indirectly (i.e., not directly but instead, after it was sent through unhosted wallet addresses) from the Pre-CryptoMixer BTC described above, further connecting the 268 post-CryptoMixer BTC addresses to the stolen funds.

²² This portion of post-CryptoMixer BTC does not account for all of the post-CryptoMixer BTC traced and identified by the FBI as stolen funds from the HTX hack.

51. These 268 BTC addresses then sent 1,123 BTC (of the total 1,145.5 BTC) to a cluster²³ of 815 BTC addresses in 250 transactions between on or about March 26, 2024, and on or about April 9, 2024. The primary address used to identify this cluster of 815 BTC addresses is bc1qtg3cyxg4yja4sut6uqkc8kmw2yyvt240srnh22 (the “bc1qtg3c Cluster”). Within the bc1qtg3c Cluster of addresses, 12 of the 815 BTC addresses received 1,123 BTC. These 12 addresses were as follows:

- bc1qtg3cyxg4yja4sut6uqkc8kmw2yyvt240srnh22
- bc1qtsjzdefpw83h9uay7afgx9zg0rnan2dkecwtnp
- bc1qg37prnadsvvqy9glhfd5hu3sg4nuw6nqs2lt7p
- bc1qeg855cyj85cmnm4qt9ax8ql5e97f2cegewquk3
- bc1q690360gpc4sq8czs7626axvx4mpvajggzvzy3m
- bc1qpkypyxtqqe7yg4xdh2c4c56hfr8fl5jsrv23
- bc1q4a9a7pt7ku46d8a9yar5pxmd8rmrdhra0nwm4
- bc1qmf7zurt59c83rv3fs2mhxkehfhmvd8fztpf3mf
- bc1q0l3qvstjmdvrfqsfexuhj7r74wvqumnhgww5me
- bc1q4zd2azutzpe9r5n4j9y87m5923cvssz60jwuv6
- bc1qhe4vvepl7kh2wshau69e0acvt6nnwwuznzycv0
- bc1q7ccmeq09lntnnl9tl6mj7uuxgrgk29mx5wwz55

²³ The FBI used virtual currency tracing software that examines public blockchain data to assess the entire history of transactions. Using deterministic methodology and clustering heuristics, the tool identifies which addresses are managed by the same entity and groups them together into a “cluster.”

52. Between March 26, 2024, and April 19, 2024, the bc1qtg3c Cluster received approximately 1,539.91 BTC and sent approximately 1,539.84 BTC, with the difference being attributed to fees of approximately .07 BTC. The bc1qtg3c Cluster sent BTC to several addresses directly and indirectly, including (but not limited to) the v764 BTC Address (a BTC cluster made up of 2,404 BTC addresses) and the 510j BTC Address, identified above as part of the “first pathway.”

53. Between on or about April 2, 2024, and on or about April 6, 2024, the bc1qtg3c Cluster sent BTC directly to the 510j BTC Address in 107 transfers, which totaled approximately 113 BTC. (This was described earlier in paragraph 37 as the first pathway.) Additionally, the 510j BTC Address received approximately 59 BTC indirectly (via multiple intermediate BTC addresses) from the bc1qtg3c Cluster, with the remaining amount coming from other funds tied to the HTX heist, totaling approximately 172 BTC (i.e., 113 BTC, plus 59 BTC, equals 172 BTC).

54. Of the 172 BTC which the 510j BTC Address received, approximately 143 BTC was sent to deposit addresses at Bitget. Bitget provided records related to the deposit addresses (i.e., account records for each account associated with any of the deposit addresses). Those deposit addresses were associated with 12 accounts/user IDs (“UIDs”). Based on review of the UID records from Bitget, law enforcement identified that the accounts were used to swap BTC into USDT on Binance’s BNB Smart Chain (BSC) via several different withdrawal addresses. From these withdrawal addresses, the funds were then consolidated into the following two BNB addresses as USDT:

- a. 0x60e33A638177d376e5e8aB7dE1eF4eb5264d1768 (“0x60e33A”)
- b. 0xb790873aC389ca77E9ef7E41704247b7C2D94EF3 (“0xb79087”)

55. The BNB address 0x60e33A received approximately \$4.6 million worth of USDT indirectly from the Bitget accounts between on or about April 3, 2024, and on or about April 6, 2024. Between on or about April 3, 2024, and on or about April 6, 2024, the BNB address 0x60e33A sent approximately \$4.816 million worth of USDT to deposit addresses at Binance, with approximately \$4.6 million being sourced from the Bitget accounts.

56. The BNB address 0xb79087 received approximately \$5.3 million worth of USDT indirectly from the Bitget accounts between on or about April 6, 2024, and on or about April 8, 2024. Between on or about April 6, 2024, and on or about April 8, 2024, the BNB address 0xb79087 sent approximately \$5.364 million worth of USDT to deposit addresses at Binance, with approximately \$5.3 million being sourced from the Bitget accounts.

Binance Referral Regarding Funds Stolen From HTX

57. On or about April 8, 2024, Binance notified law enforcement that funds tied to the theft from HTX had been laundered into USDT on TRON via “nested services” within Binance named SafePal and SWFT.pro (“SWFT”).²⁴ (SafePal and SWFT are the two virtual currency swapping services referred to above.) Swapping stolen funds across these different services increases the difficulty for investigators to trace the funds and delays the designation of the stolen funds as illicit by different blockchain, anti-money laundering and address “scoring” software solutions.

²⁴ SafePal is a wallet software that allows a user to swap one asset for another. SafePal’s service is “nested” at Binance. When a service is nested, it uses a business account at the VCE to take advantage of the liquidity available to trade one asset for another. SWFT.pro is an instant virtual currency exchange also nested at Binance.

58. On or about April 9, 2024, a Binance representative provided the FBI with SWFT records related to the BNB address 0xb79087. From these records, FBI identified swaps made between on or about April 6, 2024, and on or about April 8, 2024, from the USDT at the BNB address 0xb79087 (all of which are stolen funds from the HTX heist). Twenty-four addresses received USDT on the TRON blockchain from these swaps, 10 of which were frozen. Specifically, the USDT associated with the following 10 TRON addresses (which are 10 of the 47 Target Addresses) was voluntarily frozen by Tether pursuant to a request from law enforcement and held the following approximate values of U.S. dollars as of March 13, 2025:

Address	Token Amount (USDT)
TCxCHCndDkz2xNAaYibienjLaS8eZASaXu	258070.89625
TDskBcHHaFYsJz3QuKBSd5Q9ZxEc8dPzyH	294582.65558
TKdBozGxEFNZoAwX3rYB9gsZ373jQAYgkm	268792.09259
TLivG1x3R7Wgjlh1w7Xgz2XWcWRV82yK8c	329992.67865
TTbgnFEFEemicu1eXN3bvUE8fKbdx3di8Yb	332850.915953
TVFhKSygmJA1Qu7hrC4eavg8Esp9AoH2pu	254593.06507
TW6swH66FMUMgixHHSUUy1qDKrXZxprAKV	300333.687
TWr1G47DtfvPZ1QAH3RgV9c6SkSSe8a9hS	273021.82031
TXJeZYLn2Vr7DUwvrqceo5VRQ8b9EucxLn	285412.18378
TXKPga6EH25mexfQ7BMFGbVfGwz1rvqXVK	295697.504896
Total:	2893347.50008 (USDT)

59. Tether was not able to freeze the other 14 addresses (out of the 24 total referenced in paragraph 58) before the funds were withdrawn from the addresses and further laundered. All of the above-referenced USDT is connected to the HTX heist.

60. On or about April 13, 2024, the same Binance representative provided the FBI with SWFT records related to the BNB address 0x60e33A. From these records, FBI identified swaps made between on or about April 3, 2024, and on or about April 6, 2024, from the USDT at the BNB address 0x60e33A. Thirty-nine addresses received USDT on the TRON blockchain from these swaps; however, the funds were removed from these addresses prior to them being identified for a possible freeze.

BSC Consolidation – The Second Pathway to Additional Target Addresses

61. The Binance representative also identified for the FBI some USDT tied to the HTX heist that was located in TRON addresses. As described further below, the FBI learned through two paths of tracing – hereinafter the “Non-Wasabi Wallet Path” and the “Wasabi Wallet Path” – that the stolen funds went to SWFT, then to SafePal, then back to SWFT to be swapped into USDT in the TRON addresses of the second pathway. The total amount identified, approximately 25 percent and 75 percent of which transited the Non-Wasabi Wallet Path and the Wasabi Wallet Path, respectively, exceeded \$11 million in USDT within those TRON addresses.

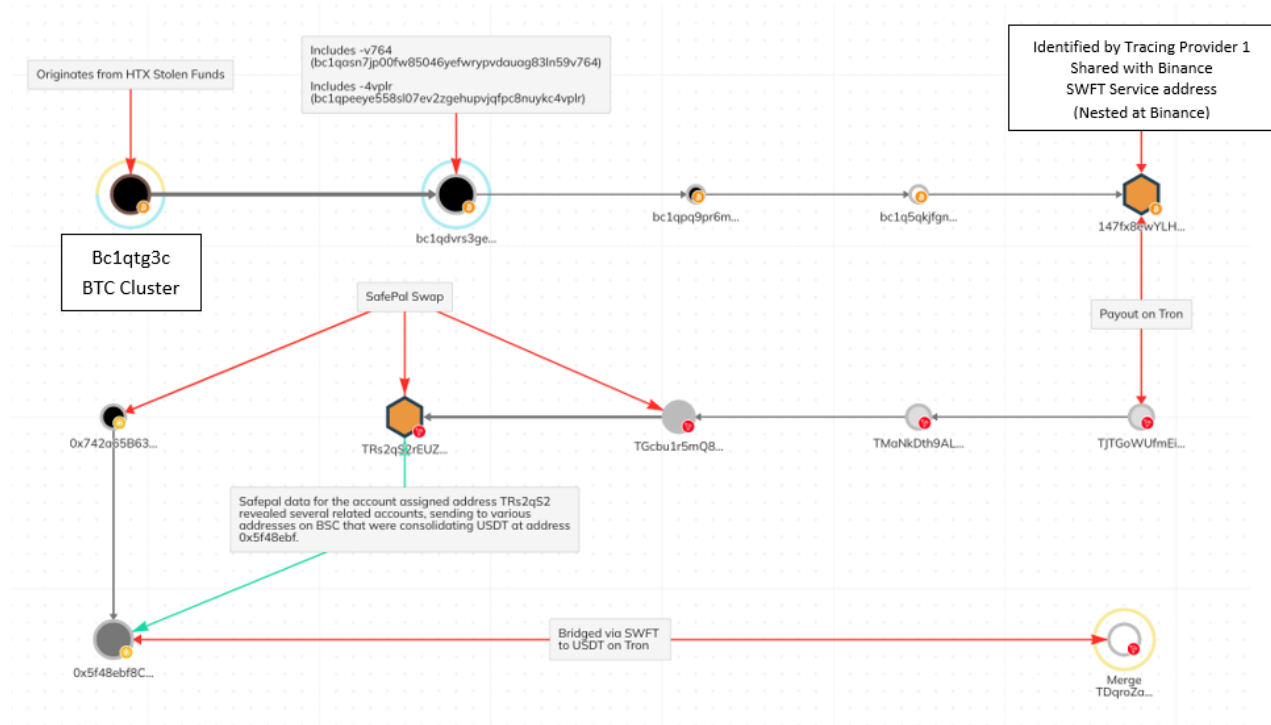
The Non-Wasabi Wallet Path

62. Binance described to the FBI its initial lead, which was an April 9, 2024 notification to Binance by Tracing Provider 1, indicating that BTC address bc1qpeeye558sl07ev2zgehupvjqfpc8nuykc4vplr (“4vplr”) may be connected to criminal activity. In sum, conspirators sent stolen BTC from BTC address 4vplr to the Binance deposit address 147fx8ewYlH9EKqk1wqqgCXWJBwdVazEQ1 (“147fx8ew”), and Tracing Provider 1 assessed that the BTC was connected to the HTX heist. The FBI reviewed the tracing provided by Tracing Provider 1 and independently confirmed the BTC transaction from 4vplr to the Binance deposit address 147fx8ew in the amount of 1.156592 BTC was connected to the HTX heist. For example, the BTC address 4vplr was one of the addresses that was part of the v764 BTC Address cluster described in paragraph 37.b. above.

63. Additionally, on or about April 9, 2024, the FBI learned from the Binance representative that Tracing Provider 2 also identified the 4vplr BTC address as a consolidation point for funds stolen from HTX.

64. The Binance representative advised that the Binance deposit address 147fx8ew (referenced above) belonged to SWFT. The Binance representative explained that the BTC from 147fx8ew was swapped to USDT on TRON at address TJTGoWUfmEi5cd9DnB1to9gyQ8uiGBnv88. The Binance representative reviewed related transactions on the TRON blockchain and identified payments going to SafePal. The Binance representative obtained records from SafePal, which revealed several additional related accounts. Specifically, the accounts were identified as being related because they were sending funds to various addresses on BSC before consolidating those funds as USDT on BSC at the address 0x5f48ebf8c6c3af25891130cee1b546ee3eee163c (“0x5f48eb”).

65. The below graphic depicts the Non-Wasabi Wallet Path flow of stolen HTX funds from BTC addresses into SWFT, where the BTC was swapped for USDT on TRON. The USDT on TRON was then swapped again via SafePal into USDT on BSC, and consolidated into 0x5f48eb as USDT.

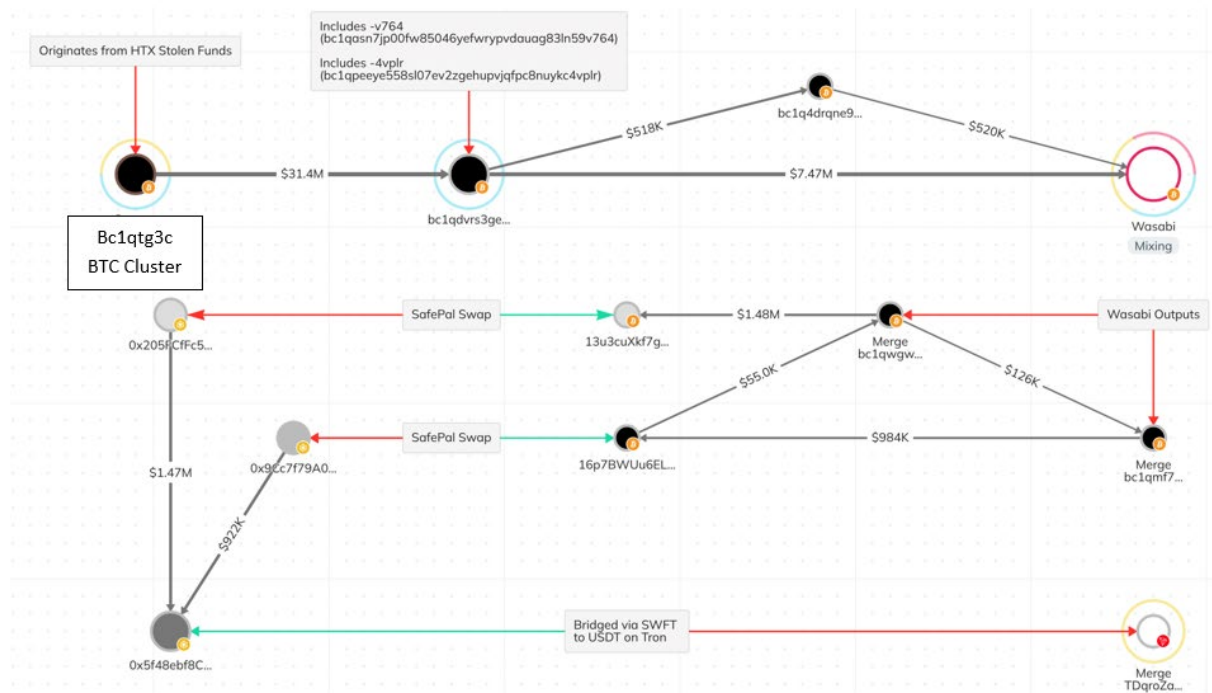


The Wasabi Wallet Path

66. By tracing funds stolen from the v764 BTC Address the FBI identified the Wasabi Wallet Path to the 0x5f48eb consolidation address. Specifically, the v764 BTC Address, which received stolen HTX funds as described above, received approximately 961 BTC and sent approximately 961 BTC. Approximately 935 BTC, or 97% of the 961 BTC received, were received from funds tied to the HTX heist. Of the 935 stolen BTC within the v764 BTC Address, approximately 723 BTC (or 75% of the total 961 BTC in v764),²⁵ were sent through Wasabi Wallet. The FBI then used tracing and records provided by private sector partners to identify the corresponding outputs from the Wasabi Wallet transacting through both SWFT and SafePal (as occurred with the Non-Wasabi Wallet Path) to the 0x5f48eb consolidation address.

67. The below graphic depicts examples of the Wasabi Wallet Path flow of stolen HTX funds from BTC addresses into Wasabi Wallet to the 0x5f48eb consolidation address on BSC and eventually to the target addresses on TRON. The Wasabi Wallet BTC outputs depicted below were swapped via SafePal into USDT on two BSC addresses which then consolidated into 0x5f48eb as USDT on BSC.

²⁵ As noted above in the Non-Wasabi Wallet Path discussion, 25% of the total BTC – *i.e.*, 212 BTC – were not sent through Wasabi Wallet.



Funds from Both the Wasabi and Non-Wasabi Wallet Paths Sent from the 0x5f48eb Consolidation Address to 37 Target Addresses

68. A review of transactions associated with the 0x5f48eb consolidation address on BSC, between on or about April 8, 2024, and on or about April 9, 2024, revealed that the address had 96 deposits, 178 withdrawals, and sent approximately \$11,321,214 of the approximately \$11,486,500 in funds it had received. From 0x5f48eb, the USDT funds on BSC were sent back to SWFT and converted to one BTC address and 37 addresses (which are 37 of the 47 Target Addresses described further below) with USDT on TRON. As described above based on both the Wasabi and Non-Wasabi Wallet Paths, the FBI assesses that all of these funds are associated with the HTX heist based on the tracing from the v764 BTC Address cluster referenced above.

69. On or about April 9, 2024, the Binance representative contacted Tether, which controls the smart contracts that create USDT tokens, to request a freeze on the 37 addresses associated with USDT on TRON that had been swapped from 0x5f48eb on BSC.

70. On or about April 9, 2024, law enforcement also requested that Tether implement a freeze on those addresses. Tether voluntarily froze the USDT held in the 37 TRON addresses (which are among the 47 Target Addresses). The following table lays out the amount of USDT frozen within each address (totaling over 11 million USDT):

Address	USDT Balance
TByiFBGTx83ptNhLiFcQJQxtnG66hs4GY3	350,261.9415
TNSn4yJoR96ZruiEWPUFLxLa1k73bVpkEh	309,727.8048
TTPjk6tJjHPWNxLwEGE4ZW8ADqGsd95zvc	273,368.621
TC6WxrikcxBQVN2vpChGwbJP9QEAPp7CoR	365,776.2585
TPjsHN4KRdaWYvUbVdMUuAeL6k7qLpTTxW	341,073.8958
TLDwBfJBRptX5nFh4rCbvhkhT5cGWbHfKR	258,569.55
TAZ6yaCrxvyVED6634GKXPntR7v6aBn7Rb	357,821.4017
TG9ZAev1Q4335uM6Gwm5df9eFUFz8e6Xgo	293,377.3125
TNBWvFHbBDjd2JzPqwSPywh8rv1C4LXjdM	296,062.4828
TDzh3ss71yCVCdKqcuQjvgaguVkJtmWSFh	332,905.0076
TN7iF7MgBSBNYFr38b3jYiCUEbREJAiWjz	304,527.7306
TXwggJm8GMcbew9J6A5myt8Gi5dxM9j8ia	277,324.7615
TDqroZaFsjWXhefyMqXthS2rvxzmWnavNZ	290,781.6479
TQ48GtajeAVJmohSBvJ4rvC9oVNZuWz57q	108,495.3847
TWYCvyCmhikaVpKx6Zf3nGWUjYbVhSGQK5	272,988.7088
TTi46bHBvmkge8NcHLfLB2587XF7GfZFs	316,238.6399
TQPQT3im4NW6MGZSQwJBXA2S6Yj2saSBSy	314,958.1253
THSBNYfGL9ja7b5xU98f8ofU2iy2gXPjSN	357,622.497

TEcpUeeiVqxmUujg2s5SAtFxcdN6P9bVB1	321,720.7763
TA9bzquWzBRKgjTPJ5gbWMWv86rYqc2yZj	258,710.773
THhic3agJ4544GuJUhuEj7bixriYfkjb8	350,460.8444
TRxz3DY9ELAoFSZyf4CuaLyEHhGYmQ8b2w	265,234.8697
TRkJ39s58grMSMqFN8YgToMxcaj95ENFxf	380,958.41
TBo6abebYprVvwABcXbpbgtgGRtmeRSrdz	421,866.4815
TZ8Rni5ixMx6SKVfY4Qz7QdZYnbtRuGCVR	262,350.6675
TEZuPBkVcNyci5WM3zrzoqX6ho4hMXgnJ6	273,487.1625
TKvUXiHow912c8HPCpwrdoz4z47NWwL9D	334,052.6693
TMYtjT2aGUtY16WKF1bkh4ssjoKFKiznqw	264,039.3413
TMunHrvu7UCM8Ro6c968SPwRx9EdLDnHW5	307,996.576
TLv2pzHZaHyY8k5nWfCE6cGGkLwdQCi3Dg	302,626.2323
TSSs4zyjdQM6L7a6v1umWmng3zzwzquSq8	263,097.5426
TK6vKMxpG6Fsrt462Nahgzg8RuzLfemnRX	339,919.0635
TU4oX4BcJgfjQUFuKAC27rKzYvqS5hNyKH	251,187.3208
TLq5jP2J2ebYghCX1mXP9gcFRfjPbcfZs	84,034.68375
TDh6QVjBaYF5kdN2qrREKydArESsG12N9h	298,349.8531
TLVSDypgdtRTd1yuGNquCLMoDhgkVrCgmo	343,898.2968
TDmBm5zUSiupoV71MXUittZqReN3h85nJc	341,730.2671
Total (USDT):	11,087,603.60335

71. In total, the Target Addresses are associated with approximately 13,980,951.103 USDT with an approximate value of \$13.977 million as of November 5, 2025. All of those funds are tied to the HTX heist.

VI. Seizure of the Defendant Property

72. On or about March 19, 2025, the FBI provided Tether with a copy of seizure warrant 25-sz-33. Tether then transferred to the United States the equivalent amount of USDT as was associated with the Target Addresses.

73. The Defendant Property is currently in the possession of the USMS.

COUNT ONE – FORFEITURE OF DEFENDANT PROPERTY

(18 U.S.C. §§ 981(a)(1)(C) & 28 U.S.C. § 2461(c))

74. Paragraphs 1 through 72 are realleged and incorporated by reference here.

75. The Defendant Property is property constituting or derived from proceeds traceable to wire fraud, wire fraud conspiracy, money laundering, money laundering conspiracy, computer fraud and computer fraud conspiracy in violation of 18 U.S.C. §§ 1343, 1349, 1956, 1030, and 371, respectively.

76. Accordingly, the Defendant Property is subject to forfeiture to the United States under 18 U.S.C. § 981(a)(1)(C) and 28 U.S.C. § 2461(c).

COUNT TWO – FORFEITURE OF DEFENDANT PROPERTY

(18 U.S.C. §§ 981(a)(1)(A))

77. Paragraphs 1 through 72 are realleged and incorporated by reference here.

78. The Defendant Property is property involved in a transaction or attempted transaction in violation of 18 U.S.C. §§ 1956(a)(1)(A)(i), 1956(a)(1)(C)(i), and 1956(h), that is, a conspiracy to conduct or attempt to conduct financial transactions involving the proceeds of specified unlawful

activity, to wit, wire fraud, computer fraud, and conspiracy to commit these offenses, with the intent to promote the carrying on of specified unlawful activity, knowing that the transaction was designed in whole and in part to conceal and disguise the nature, location, source, ownership, and control of the proceeds of specified unlawful activity, and knowing that the property involved in the financial transaction represented the proceeds of some form of unlawful activity.

79. Accordingly, the Defendant Property is subject to forfeiture to the United States under 18 U.S.C. § 981(a)(1)(A).

PRAYER FOR RELIEF

80. WHEREFORE, the United States of America prays that notice issue on the Defendant Property as described above; that due notice be given to all parties to appear and show cause why the forfeiture should not be decreed; that this Honorable Court issue a warrant of arrest *in rem* according to law; that judgment be entered declaring that the Defendant Property be forfeited to the United States of America for disposition according to law; and that the United States of America be granted such other relief as this Court may deem just and proper.

November 14, 2025
Washington, D.C.

Respectfully submitted,

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/s / Rick Blaylock, Jr.
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VERIFICATION

I, Justin M. Vallese, a Special Agent with the Federal Bureau of Investigation, declare under penalty of perjury, pursuant to 28 U.S.C. § 1746, that the foregoing Verified Complaint for Forfeiture *in rem* is based upon reports and information known to me and/or furnished to me by other law enforcement representatives and that everything represented herein is true and correct.

Executed on this 14th day of November 2025.


Justin M. Vallese
Special Agent
Federal Bureau of Investigation